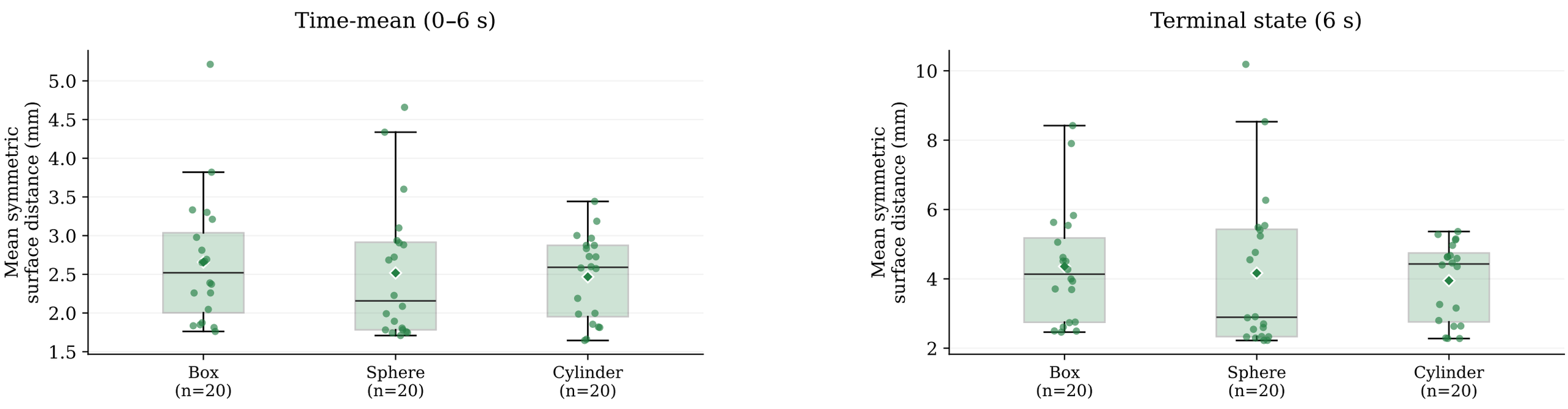

Figure 2: Frozen object-model checkpoint 11,000. Rows show the mean and 95th-percentile symmetric surface distance. Left column: time-mean over 0–6 s; right column: terminal state at 6 s. Distances use area-uniform surface samples without alignment. The 95th percentile is over surface distances within a frame, not across trajectories. Sand surfaces use a 0.02 m measurement grid. Each dot is one trajectory summary; boxes show the median and interquartile range, whiskers extend to 1.5 IQR, diamonds mark means. Twenty trajectories per shape unless labeled otherwise. Time means use available samples; empty particle regions are not assigned zero error.